On the conversion of module representations for higher dimensional supersingular isogenies

Joint work with A. Page and D. Robert

Julien Soumier

Isogeny Club, 8th Season

Motivations

Useful tools:

- Deuring correspondence,
- Ideal class group action and Clapoti(s).

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In the following, $\mathcal{O}$ denotes a maximal quaternionic order in $\mathcal{B}_{p,\infty}$.

[1] [Module action ; Robert 2024]

Contributions[1]:

Theorem: PIP (GRH)

Let I be an ideal in $\mathcal{M}_g(\mathcal{O})$, $g \geq 2$.
We can compute a generator of I.

We also provide a Sage implementation.

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- We can compute isomorphisms $A \simeq B$ of *maximal* abelian varieties.

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Contents

1. **Abelian varieties and Hermitian modules**
2. Principal Ideal Problem
3. HD-Clapotis and applications
4. Switching between the different representations

Background

Abelian variety

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In dimension $g > 1$:
$E_1 \times \cdots \times E_g$, $\mathrm{Jac}(C)$, ...
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Maximal abelian variety[1]

Definition:

Let $A/\mathbb{F}_{p^2}$ be an abelian variety of dimension g. A is said to be maximal when:

$$\mathrm{Card}(A(\mathbb{F}_{p^2})) = (p+1)^{2g}.$$

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In the following, E is a fixed maximal elliptic curve with $\mathrm{End}(E) \simeq \mathcal{O}$.

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- *[Montessinos ; 2026]* solves the case for any principally polarized abelian surface,
  $\Rightarrow$ with known IKO matrix.
- *[Page, Robert, S. ; 2026]* solves the general case,
  $\Rightarrow$ given the module represention of A.

Hermitian modules[1], definitions

Fix M a finitely presented torsion-free module of rank g, over $\mathcal{O}$.

[1] Most of the results holds for $\mathcal{O}$ an imaginary quadratic order

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Hermitian form, $H_M : M \times M \to \mathcal{O}$:

A sesquilinear form such that:

- $H_M(x, y) = \overline{H_M(y, x)}$,
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Such a couple (M, H_M) is called an *Hermitian module*.

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Hermitian modules, examples

Base module

$(\mathcal{O}, H_0)$, where
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(M, H_M) is *unimodular*, when H_M induces an isomorphism $M \simeq M^\vee$.

(I, H_0) is not unimodular, but (I, H_I) is, where $H_I := \frac{1}{\mathrm{Nrd}(I)} H_0$.

Link with abelian varieties

Fact: There is a correspondence between:

Maximal abelian varieties. $\rightleftarrows$ good $\mathcal{O}$-modules M.

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Maximal abelian variety $A/\mathbb{F}_{p^2}$ of dimension g	$\mathcal{O}$-module $M \simeq \mathrm{Hom}_{\mathbb{F}_{p^2}}(A, E)$ of rank g
The dual abelian variety $\widehat{A}$	The dual $M^\vee$
Principal polarization $\lambda_A \colon A \to \widehat{A}$	Unimodular Hermitian form H_M

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iii) The (principally polarized) abelian variety associated with $(\bigoplus_{i \leq g} I_i, \bigoplus_{i \leq g} H_{I_i})$ is $(\prod_{i \leq g} E_{I_i}, \bigotimes_{i \leq g} \lambda_i)$.

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Remark:

A (projective) rank 1 $\mathcal{O}$-module carries less information that an (invertible) ideal of $\mathcal{O}$. We do not need to remember the inclusion $I \hookrightarrow \mathcal{O}$.

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Standard representation

Theorem: Deligne-Ogus-Shioda for modules

Let M be a good $\mathcal{O}$-module of rank ($g \geq 2$).
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Remark:

Looking at the Gram matrices for $g = 2$, we recover the matrices from $KLPT^2$, *[Castryck, Decru, Kutas, Laval, Petit, Ti ; 2025]*.

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Finding $\mathcal{O}$-basis of modules is equivalent to principalizing ideals in $\mathcal{M}_g(\mathcal{O})$[1].

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For any right ideal $I \subset \mathcal{M}_g(\mathcal{O})$, find a generator $\alpha \in I$, $I = \alpha\mathcal{M}_g(\mathcal{O})$.

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General strategy

By the local-global principle, it suffices to find $\alpha \in \mathcal{M}_g(\mathcal{O})$ such that:

$$I \otimes \mathbb{Z}_q = \alpha\mathcal{M}_g(\mathcal{O}) \otimes \mathbb{Z}_q \text{ for every prime } q.$$

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$\implies$ Work in $\mathcal{M}_g(\mathcal{O}) \otimes \mathbb{Z}_\ell \simeq \mathcal{M}_{2g}(\mathbb{Z}_\ell)$.

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Ingredients:

- KLPT solves the case $g = 1$,
- Conclude by $\mathcal{O}$-linear algebra.

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Thus $I = \alpha_1\alpha_2\mathcal{M}_g(\mathcal{O})$.

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- We used GRH in the first step to call KLPT efficiently.
- We reduced[1] to the case where $\mathcal{O}$ has an element $u \in \mathcal{O}$ of norm $\mathrm{Nrd}(u) = \ell$.

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`https://gitlab.inria.fr/module-representation/PIP-quaternion-mat`

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Maps between modules

Geometric side

An isogeny is a *surjective morphism* $\varphi : A \twoheadrightarrow B$ with *finite kernel*.

$\varphi : (A, \lambda_A) \twoheadrightarrow (B, \lambda_B)$ is an n-isogeny, if $\varphi^* \lambda_B = [n]\lambda_A$.

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Module side

A module isogeny is an *injective morphism* $\psi : M_B \hookrightarrow M_A$ with *finite cokernel*.

$\psi : (M_B, H_B) \hookrightarrow (M_A, H_A)$ is an n-similitude, if $\psi^* H_A = [n]H_B$.

Maps between modules, examples

Similitudes:

i) Let $x \in (M, H_M)$. The action $\lambda \mapsto \lambda \cdot x$, induces an isomorphism $(\mathcal{O}, H_0) \simeq (\mathcal{O}x, H_M/q_M(x))$.

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General idea:

Computing n-isogenies $A \to B \iff$ Constructing submodules $M_B \hookrightarrow M_A$.

The 1-dimensional Clapoti(s)

Objective of Clapoti(s):

Let E be an elliptic curve oriented by R quadratic imaginary. Let $\mathfrak{a}$ be an ideal of R.
We want to compute the action $E_{\mathfrak{a}} := \mathfrak{a} \cdot E$.

[1] From Clapoti(s), *[Page, Robert ; 2023]*.

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Ingredients[1]:

- Compute two ideals $\mathfrak{b}$ and $\mathfrak{c}$ of coprime norm (and equivalent to $\mathfrak{a}$).
- Use a Kani diagram to efficiently compute an isogeny.

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Statment of our version of HD-clapotis

Theorem: HD-Clapotis

Given (M, H_M), we can compute the action $(M, H_M) \cdot E = (A, \lambda_A)$.

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Idea:

By the same strategy as Clapoti(s)[1], if we have a double path[2] we can compute (A, λ_A) along with the corresponding coprime isogenies $(A, \lambda_A) \to (E^g, \lambda^g)$.

[1] See *[Module action ; Robert 2024], Proposition 4.30.*

[2] Double paths are also in *[SQISignHD ; Dartois, Leroux, Robert, Wesolowski 2023]*

Constructing coprime orthogonal full submodules

Theorem: Module double path

We can efficiently compute two coprime D_i-similitudes $\psi_i : (\mathcal{O}^g, H_0^g) \to (M, H_M)$.

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Consequence: (GRH)

Given (M, H_M) representing (A, λ_A), we can recover an isomorphism $A \simeq E^g$.

Application: Computing isomorphism $A \simeq E^g$

First we compute (polarized) coprime double-path ψ_i.

$$(\mathcal{O}^g, H_0^g) \overset{\psi_1}{\underset{\psi_2}{\rightrightarrows}} (M, H_M)$$

Application: Computing isomorphism $A \simeq E^g$

First we compute (polarized) coprime double-path ψ_i.
Then an (unpolarized) isomorphism ψ thanks to our PIP solution.

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We recover the isogenies φ_i (resp. φ'_i) corresponding to ψ_i (resp. $\psi \circ \psi_i$).

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$$\varphi(P) = d_1^{-1} \widehat{\varphi_1}(\varphi'_1(P)).$$

Contents

1. Abelian varieties and Hermitian modules

2. Principal Ideal Problem

3. HD-Clapotis and applications

4. **Switching between the different representations**

Complete representation, definition

Complete representation of $(\mathrm{End}(A), r_A)$

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Since $A \simeq E^g$ we can take $R = \mathcal{M}_g(\mathcal{O}) \simeq \mathrm{End}(E^g) \simeq \mathrm{End}(A)$.

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Rosati r_S on R

If $\psi : (\mathcal{O}^g, S) \simeq (M, H_M)$, where
$S \in \mathcal{M}_g(\mathcal{O})$ is seen as a Gram matrix.
Define $r_S : N \mapsto SN^*S^{-1}$.

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General for $\mathrm{Hom}(A_1, B_1)$

$$\begin{tikzcd} A_1 \arrow[r, dashed, "abstract"] \arrow[d, shift right] \arrow[d, shift left] & B_1 \arrow[d, shift right] \arrow[d, shift left] \\ A_2 \arrow[r, "complete"'] & B_2 \end{tikzcd}$$

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Solution given in *[Montessinos, 2026]*:
$\Longrightarrow$ Remember that M induces a structure of $\mathbb{F}_{p^2}$-algebra on $\mathcal{M}_g(\mathbb{F}_{p^2}) \simeq \mathrm{End}(M/\mathfrak{p}M)$.

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Thank you for your attention!!